1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist highlights a recurring paradox — markers of immune infiltration (Immunoscore/TILs) and noninvasive methylation signatures (SDC2/TFPI2) both predict NMIBC outcomes, yet some TIL-rich tumors still progress. Why does apparent immune infiltration sometimes fail to translate into durable control (longer progression-free interval)?

Working Hypothesis

In NMIBC, high TIL density predicts prolonged PFI only when invasion-front stromal signaling remains intact; promoter methylation of SDC2 and TFPI2 disrupts stromal proteoglycan signaling and promotes ECM remodeling at the invasion front, creating an immune-excluded microenvironment. Specifically, SDC2/TFPI2 promoter methylation leads to reduced stromal signaling and increased invasion-front remodeling (e.g., via MMP activity), which mislocalizes CD8+ T cells away from tumor nests and impairs their cytotoxic function, resulting in shorter PFI despite apparent immune infiltration. This hypothesis yields testable predictions: (i) among TIL-high NMIBC tumors, those with SDC2/TFPI2 promoter methylation will have shorter PFI than unmethylated counterparts; (ii) spatial analyses will show T cells predominantly in stroma rather than tumor nests in methylated tumors; (iii) experimentally inducing demethylation or preventing SDC2/TFPI2 silencing will restore invasion-front signaling, enhance T cell–tumor contact, and prolong PFI. Feasible approaches include multi-omics cohort analyses (TIL density, methylation status, PFI), spatial transcriptomics/immunofluorescence for TIL localization, and in vitro/in vivo models employing targeted methylation/demethylation of SDC2 and TFPI2 to assess ECM remodeling and T cell behavior.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	SDC2, TFPI2
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

SDC2: SDC2/TFPI2 stool methylation (core path 3) because epigenetic silencing of stromal/ECM regulators could mechanistically link tumor epigenotype to invasion-front remodeling and immune dysfunction.

TFPI2: TFPI2 stool methylation (core path 3) because epigenetic silencing of stromal/ECM regulators could mechanistically link tumor epigenotype to invasion-front remodeling and immune dysfunction.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	SOX2	Supporting modifier from knowledge graph
Secondary	IGF2	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph
Secondary	MSH6	Supporting modifier from knowledge graph
Secondary	CD44	Supporting modifier from knowledge graph
Secondary	CTLA4	Supporting modifier from knowledge graph
Secondary	LRPPRC	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: NMIBC outcomes depend on invasion-front biology and immune contexture; immune infiltration alone (TIL density) often predicts better prognosis but can be overridden by immune-excluded microenvironments;

Prediction 2: SDC2 is a cell-surface proteoglycan involved in cell–ECM interactions, while TFPI2 inhibits proteolytic remodeling; promoter methylation likely downregulates these genes, promoting ECM/invasion-front remodeling and potentially reducing T cell access to tumor nests;

Prediction 3: integrating TIL density with epigenetic markers of invasion-front remodeling provides a testable axis (epigenetic–stroma–immune) that could explain paradoxical PFI outcomes and identify patients who remain at risk despite apparent immune infiltration.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.
- Multi-modal validation pipeline: computational -> spatial -> functional, providing mechanistic depth beyond association.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- SOX2: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- IGF2: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.